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Discrete Event Simulation for Healthcare Organizations: A Tool for Decision Making

Eric Hamrock, senior project administrator, operations integration, Johns Hopkins Health System, Baltimore, Maryland; Kerrie Paige, PhD, president, NovaSim, Bellingham, Washington; Jennifer Parks, assistant director, case mix information management, Johns Hopkins Health System; James Scheulen, chief administrative officer, emergency medicine, Johns Hopkins Hospital; and Scott Levin, PhD, associate professor, emergency medicine, Johns Hopkins University School of Medicine, and operations integration, Johns Hopkins Health System

EXECUTIVE SUMMARY

Healthcare organizations face challenges in efficiently accommodating increased patient demand with limited resources and capacity. The modern reimbursement environment prioritizes the maximization of operational efficiency and the reduction of unnecessary costs (i.e., waste) while maintaining or improving quality. As healthcare organizations adapt, significant pressures are placed on leaders to make difficult operational and budgetary decisions. In lieu of hard data, decision makers often base these decisions on subjective information. Discrete event simulation (DES), a computerized method of imitating the operation of a real-world system (e.g., healthcare delivery facility) over time, can provide decision makers with an evidence-based tool to develop and objectively vet operational solutions prior to implementation.

DES in healthcare commonly focuses on (1) improving patient flow, (2) managing bed capacity, (3) scheduling staff, (4) managing patient admission and scheduling procedures, and (5) using ancillary resources (e.g., labs, pharmacies).

This article describes applicable scenarios, outlines DES concepts, and describes the steps required for development. An original DES model developed to examine crowding and patient flow for staffing decision making at an urban academic emergency department serves as a practical example.

For more information about the concepts in this article, contact Mr. Hamrock at ehamroc1@jhmi.edu.

INTRODUCTION

Discrete event simulation (DES) is a type of computer simulation that imitates the operation of a real-world system. This type of virtualization provides users with broad capability to examine performance measures for potential operational changes and to plan for new or changing facilities. Performance measures comprise the output of the DES model and commonly include patient throughput, timeliness of care (i.e., extent of waiting), and resource utilization (e.g., bed occupancy, magnetic resonance imaging [MRI] utilization, nurse utilization). DES allows administrators and managers to analyze the interaction of management priorities and the impact of operational decisions. For example, DES is able to demonstrate the negative impact that high rates of bed occupancy (i.e., resource utilization) may have on patient waiting and throughput measures in the emergency department (ED). The accuracy and applicability of DES design depend on the performance measures targeted for examination and improvement.

DES has been shown to be effective in many healthcare settings. Some models of outpatient clinics aim to improve patient flow, reduce wait times, maximize staff utilization, and accomplish other gains in efficiency. These outpatient models are tested through changes to patient scheduling, patient routing, and internal work processes (Smith & Warner, 1971; Rising, Baron, & Averill, 1973; Bailey, 1952; Smith, Schroer, & Shannon, 1979; Williams, Covert, & Steele, 1967; Parks, Engblom, Hamrock, Satjapot, & Levin, 2011; Rohleder, Lewkonia, Bischak, Duffy, & Hendijani, 2011). Bed capacity, bed allocation, and length of stay (LOS) have been major points of focus of DES within inpatient settings (Lowery, 1992; Lowery & Martin,1992; Dumas, 1984, 1985; Cohen, Hershey, & Weiss, 1980; Zilm, Arch, & Hollis, 1983; Bagust, Place, & Posnett, 1999; El-Darzi, Vasilakis, Chaussalet, & Millard, 1998). For example, DES was used within a hospital to determine how changes in bed allocation and discharge patterns affected hospital occupancy rates (Zhu, 2011; Dumas, 1985).

DES models have also been developed to improve bed management practices and to plan for increases in admissions (e.g., surgical service expansion) (Levin et al., 2008; Levin, Dittus, Aronsky, Wienger, & France, 2011). This tool is used to develop block scheduling algorithms and examine patient flow in surgical centers (Murphy & Sigal, 1985; Fitzpatrick, Baker, & Dave, 1993).

Some DES models focused on staffing typically aim to determine the impact of staffing changes on performance measures or to design staff schedules (Vemuri, 1984; McHugh, 1989; Hashimoto & Bell, 1996; Wilt & Goddin, 1989; Rossetti, Trzcinski, & Syverud, 1999; Jun, Jacobson, & Swisher, 1999). A study by Dittus, Klein, DeBrota, Dame, & Fitzgerald (1996) used DES to develop optimal resident physician work schedules to meet hospital needs and to mitigate negative effects on both residents and the quality of care they provide caused by changes in sleep schedules and daily work activities.

Ancillary laboratories, radiology facilities, and pharmacies have used DES models to improve internal processes and timeliness of service delivery (Day,

Li, Ingolfsson, & Ravi, 2010; Tan, Chua, Yong, & Wu, 2009; Ishimoto, Ishimitsu, Koshiro, & Hirose, 1990; Reynolds et al., 2011; Coelli, Ferreira, Almeida, & Pereira, 2007). One pharmacy used DES to redesign internal work processes, prioritization systems, and staffing patterns in response to changes in workload (Reynolds et al., 2011). Uses of DES also include determining ancillary service impact on patient throughput in other care areas (i.e., inpatient, outpatient).

Similarly, the ED has been a significant focus of DES studies in healthcare, where excessive crowding poses challenges to delivering safe and timely care (Committee on the Future of Emergency Care, 2007). Several studies using DES within the ED examined patient flow and patient routing through care areas on the basis of clinical prioritization (e.g., triaging, fast tracking) (Coats & Michalis, 2001; Kirtland, Lockwood, Poisker, Stamp, & Wolfe, 1995; Garcia, Centeno, Rivera, & DeCario, 1995). Optimal bed capacity and staff sizing may also be targeted (Baesler & DaCosta, 2003; Draeger, 1992; Gabaeff & Lennon, 1991; Carter, O'Brien-Pallas, Blake, McGillis, & Zhu, 1993; Badri & Hollingsworth, 1993; Klafehn, & Owens, 1987; Bagust et al., 1999), and reducing ED LOS is another common emphasis (Connelly & Bair, 2004; McGuire, 1994), as illustrated in the section titled "Simulation Development with Practical Example."

DISCRETE EVENT SIMULATION CONCEPTS

Standard inputs to DES models include *entities*—what flow through the system—resources, locations, arrival rates, service times, and processing logic. Patients are the most common entities modeled in healthcare DES applications; however, lab specimens (e.g., blood), patient charts, and supplies are also examples of entities. *Resources* process entities through the system and include human resources (e.g., nurses, physicians, lab technicians, pharmacists) and equipment (e.g., MRI machines, lab analyzers, electrocardiograph machines). *Locations* are physical areas where resources process entities, such as treatment areas, registration, computer workstations, and lab and radiology facilities. Locations are often designed using facility layouts and scaled within the DES model. *Arrival rates* define the rate at which entities arrive at specific locations; examples might be the frequency of patients arriving at the ED or a scheduled clinic visit. *Service times* define the time needed for resources to process entities at set locations.

ED and clinic arrival rates, which define patient volume and service times, are entered into the DES in the form of probability distributions. This process roots the model in probability to account for variability in operations. For example, patients arriving to the ED may take an average of 7 minutes to be triaged, with times commonly following a Weibull distribution. An example of a Weibull probability distribution of ED triage times is shown in Figure 1. Service times in many industries, including healthcare, often follow specific probability distributions, such as Weibull, exponential, lognormal, and Erlang. Similarly, arrival times are inputted as probability distributions capturing the time between consecutive arrivals,

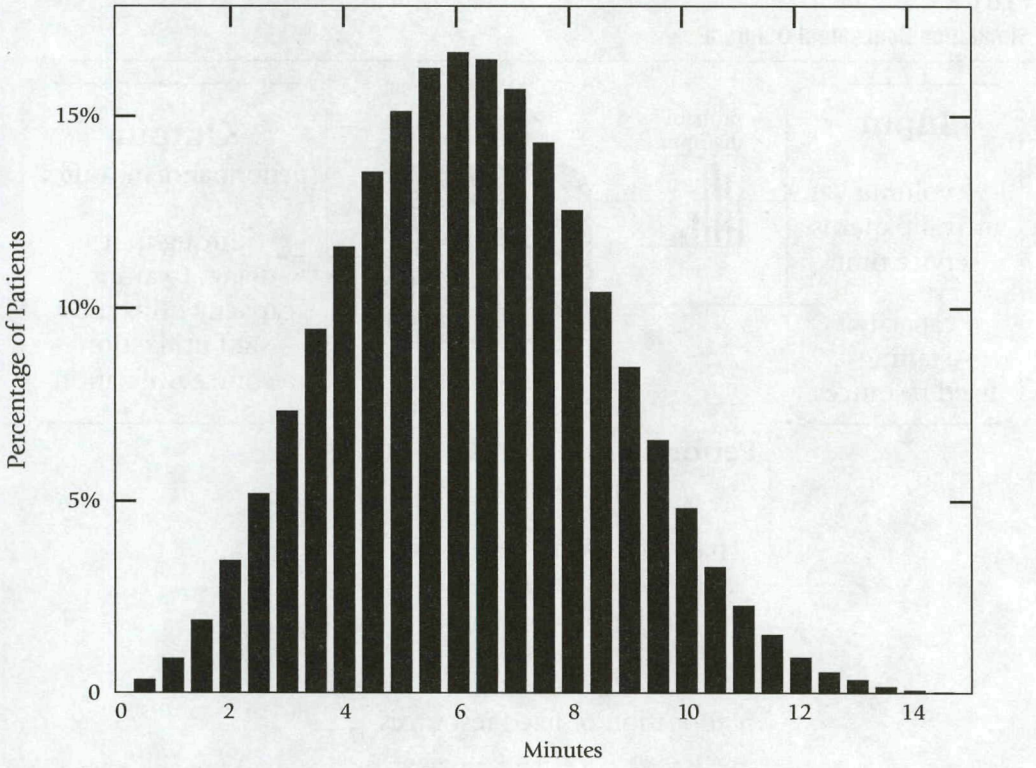
known as the *interarrival rate.* Interarrival times are most often exponentially distributed. This rate varies drastically over the course of the day and week. For example, ED patients arrive much more frequently at 6:00 p.m. (18:00) on Tuesdays than at 6:00 a.m. (06:00) on Thursdays. The arrival rate may even change by season to account for volume changes associated with influenza or seasonal shifts in populations.

Simulation time is typically broken down into pieces (i.e., day of week and hour of day), with separate arrival rates applied to each piece to reflect the daily and weekly patterns of arrivals.

Processing logic is the glue that holds all the model elements together. Processing logic determines the rules for how entities flow through the system and how resources and entities interact (i.e., discrete events). Labels are often assigned to entities to differentiate their characteristics and may change, on the basis of logic, as the entities flow through the system.

Conceptually, a DES model may be used to explore performance in a variety of ways. Figure 2 outlines the process of examining different scenarios within the model. Any elements within the model may be altered to determine the effects

FIGURE 1
Weibull Distribution of ED Triage Times

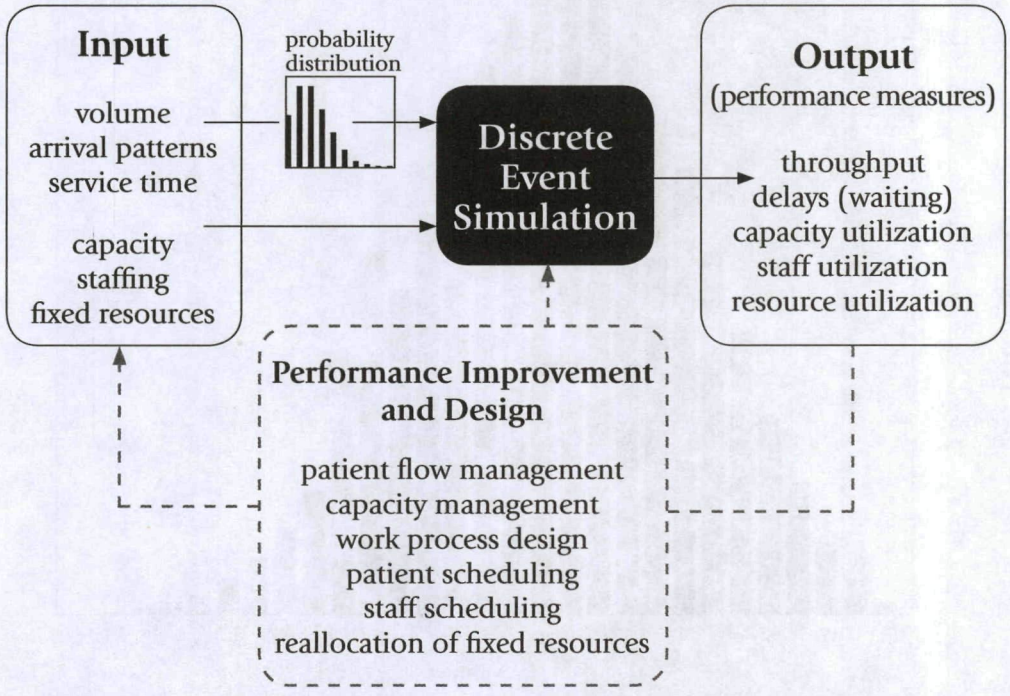


on output (i.e., performance measures). This option not only allows users to test a wide array of changes but also supports development of new strategies. Specific objectives of DES in healthcare include (1) improving patient flow, (2) managing bed capacity, (3) streamlining patient admissions and scheduling procedures, (4) improving staff scheduling, and (5) maximizing the use of ancillary resources (e.g., labs, pharmacies).

VALUE OF SIMULATION IN HEALTHCARE

New challenges are prompting healthcare organizations to change the way they deliver care. Economic constraints and service-based targets have placed pressure on healthcare decision makers to find the most efficient and effective ways to serve patients (Jack, Powers, & Nowak, 2005; ASQ, 2011). Even the most viable organizations must improve operational efficiency and reduce unnecessary costs (i.e., waste) while maintaining or improving quality of care (Story, 2011; AHRQ, 2008). Reducing risk of failure or unintended consequences along the way is desirable. Although these objectives are clear, determining the best pathway toward achieving high-impact (sustainable) improvement is often complex. Frequently, healthcare decision makers use subjective information from frontline staff, providers, and other stakeholders to make strategic improvement decisions. Change attempts—whether to structures (e.g.,

FIGURE 2
Simulation Conceptual Diagram



change in floor plan or layout) or to processes—may prove costly in terms of time and capital (Hwang, Lee, & Shin, 2011). Compared to change initiatives, tools such as DES provide a low-risk, lower-cost method to develop improvement strategies, test assumptions, and observe potential outcomes of decisions prior to implementation.

DES is valuable to decision makers because it shows not only simulation outputs of individual care processes but also how those processes interact as a whole in the system, providing a macro-level view (Story, 2011). Much of the ultimate value of a DES comes from creating the model. Defining operations, mapping care processes, and gathering data in a structured manner are valuable steps in understanding current operations and triggering ideas for improvement. DES is a powerful mathematical tool that facilitates systematic thinking around operations and objectives. Understanding how components (i.e., patients and resources) interact and the trade-offs involved in an increasingly complex and disaggregated system is often difficult to characterize without the use of such a tool. A 2005 joint report by the Institute of Medicine (IOM) and National Academy of Engineering highlights DES as a particularly valuable tool to aid in understanding how healthcare systems operate, meet objectives, and can be improved (Reid, Compton, Grossman, & Fanjiang, 2005).

The level of success achieved in attempting organizational change depends on many factors, including organizational culture, perceived benefits from the change, and outcomes from change initiatives in the past. One of the most important factors in an organizational change initiative is the perceived risk of the change or the decision to be made (Greve, 1998). Although DES does not eliminate the risk inherent in every decision, it allows risk to be forecast and accounted for.

Furthermore, many decisions within the healthcare industry require the buy-in and input of physicians, who are often the most influential stakeholders. One study of 10 hospital CEOs and physicians determined that physicians take data into account in their decision making and respect firmness, openness, and honesty in discussions about decisions to be made (Sheldon, 2006). DES allows physicians to participate in developing the model and to see the outputs of that model. An added benefit is that their participation increases transparency because the decision-making process relies on objective, data-driven output instead of subjective opinion.

A key, and often underappreciated, value of DES is that it helps decision makers understand variability within the system modeled. Many sources of variability in healthcare challenge hospital staff's and administrators' ability to efficiently manage resources. This variability may be a source of system defects or breakdown (Story, 2011; Litvak & Long, 2000; McManus et al., 2003; Litvak et al., 2005). Variability in patient flow stems from variation in arrival patterns (e.g., natural arrivals at an ED, scheduled patients within a clinic, electively scheduled procedures in an operating room) and patients' LOS. Professional variability consists of variation in providers' ability to treat patients

(e.g., timeliness) within the healthcare setting. Artificial (i.e., man-made) variability is associated with setting work processes, scheduling staff, and establishing general practices (e.g., staff–information technology interaction and documentation). DES is most valuable in targeting improvements in managing and eliminating artificial variability, and this is the area in which healthcare organizations have the most control and are most amenable to operational improvements.

SIMULATION DEVELOPMENT WITH PRACTICAL EXAMPLE

The process of simulation development from conceptualization to results and recommendations is step-wise (outlined in Figure 3). These steps are described here and demonstrated later using a practical example of DES created for an urban, academic ED.

Specifying Objectives

The first step of model development is to determine the specific objectives (see Step 1 in Figure 3). What information are administrators looking to gain? DES is capable of examining many aspects of operations; however, it is important to focus on no more than two or three specific objectives. A narrowed focus reduces the complexity of the model and creates a feasible (in terms of time and resources) set of tasks required to complete the simulation. The objective of the DES created for this study was to determine how increasing patient throughput (i.e., reducing patient LOS) would change bed occupancy rates, required staffing levels, and patient wait times. The hypothesis was that improving throughput would result in reductions in ED occupancy, allowing for proportional adjustments in staffing levels over time.

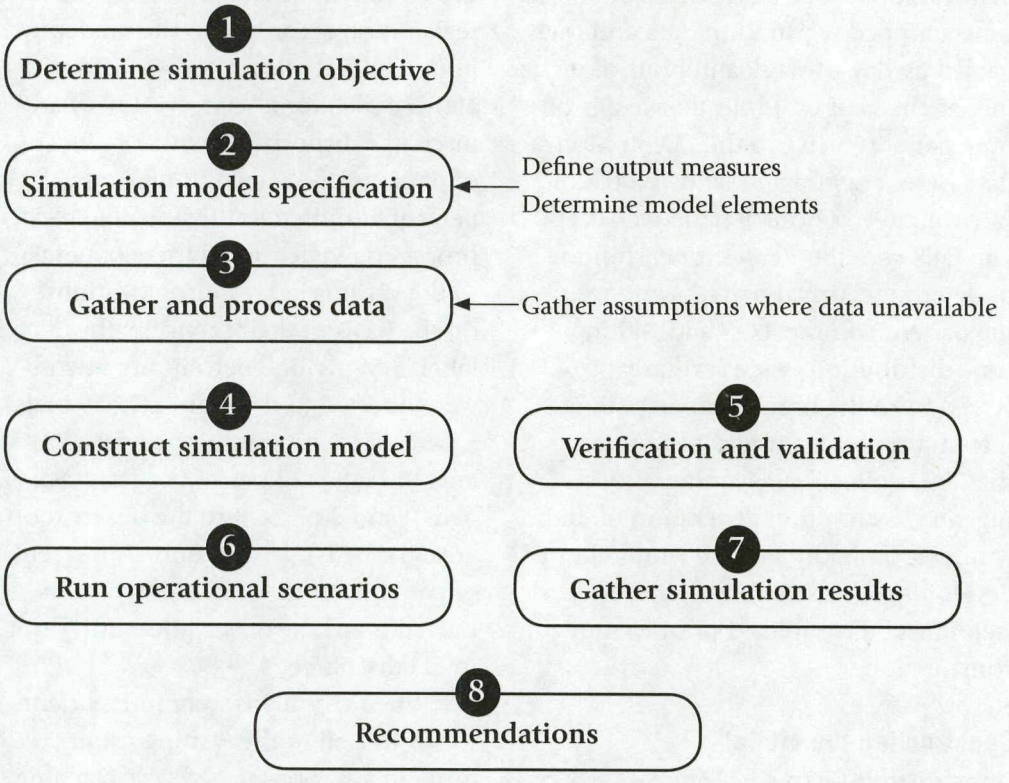
Defining Objectives

Defined objectives, including the exact output measures, facilitate DES model specification (Step 2 in Figure 3). In this example, ED bed occupancy (i.e., patient census) was the primary output measure. In addition, patient wait times and waiting room occupancy were captured outputs. Next, the elements included in the model must be specified. Patients were the entities, and treatment areas (i.e., beds), registration areas, and the waiting room were resources included in the model. The DES only captured patient flow with respect to bed utilization. No staffing resources were explicitly modeled. Patient arrival rates, LOS probability distributions, and processing logic defining patient routing comprise the remaining elements modeled.

Gathering Data

Data must then be gathered for each model element (Step 3 in Figure 3). During this information-gathering phase, it is important to assess the level of detail needed to meet objectives. Inadequate detail may lead to an inaccurate reflection of the real system; however, the marginal utility of more detail may not justify the additional data collection effort required. Overspecification (i.e., too much detail) may be too costly and time intensive for practical needs (Young, Eatock, Jahangirian, Naseer, & Lilford, 2009). In addition, data may be insufficient to describe a model element. This situation calls for assumptions to

FIGURE 3
Steps in Simulation Development



be made and entered into the model. This is a common practice whereby the DES integrates actual data with a set of assumptions. Document these assumptions and how they were derived, and gauge the influence they exert on model output.

Data collection for the ED simulation involved extracting all patient flow information for one quarter from September 1, 2009, to January 1, 2010. These data were captured retrospectively via the electronic medical record and patient management (whiteboard) systems operating within the ED. Patient information included time of registration (arrival), time to bed and bed type, time of disposition, and time of discharge or admission to the hospital. The data were then cleaned and processed to fit as input within the DES. Cleaning data to produce coherent information may be cumbersome, and users may be tempted to skip this step, but it is critical in completing the process.

Outliers have the potential to skew simulation input. For example, patients cared for in the ED for 5 minutes or for 30 days are outliers. A simple way to account for the outliers is to use only data that summarize 95 percent of the population, omitting the lowest and highest 2.5 percent. Once a clean set of data exists, it must be further

processed to operate within the DES. Input arrival rates (i.e., interarrival time distributions) were processed from time-stamped registration data and then parsed by day-of-week and hour-of-day increments. For example, an average of nine patients arrive to the ED on Mondays between 9:00 a.m. and 10:00 a.m. (arrival rate = 1 hour/9 patients = 0.11). The DES uses these rates for each hour to determine arrival patterns and resulting patient volume. LOS and waiting time distributions were similarly processed from the time-stamped patient information. It is important to note that data collection, cleaning, processing, and assumption generation often comprise a majority of the workload in developing a simulation model. This step must be completed prior to model construction.

Constructing the Model

Construction (Step 4 in Figure 3) involves taking all the data collected and integrating (i.e., programming) it within a DES modeling framework. Examples of software packages used in healthcare applications include MedModel, Arena, SIMUL8, FlexSim, and Mathworks SimEvents. Software packages range in cost and the technical skill needed for development. Development environments may require a large degree of direct programming (code writing) or be designed with a graphical user interface that allows for easier use with less technical skill. Typically a trade-off is made between ease of use and power and flexibility; however, many software packages adapt to both beginning (i.e., graphical user interface) and more advanced (i.e., direct programming) users.

SIMUL8 was used to construct the DES described here. In the model, the design called for patients to arrive at registration according to the predefined hourly arrival rates. They were immediately labeled for a service pathway and discharge disposition, with 79 percent of patients discharged home and 21 percent admitted to the hospital. Next, processing logic routed these patients to the various ED resource locations on the basis of their service pathway label. Seventy-one percent of patients were directed to the main ED (25 beds), 5 percent to the psychiatric treatment area (6 beds), 22 percent to fast track (7 beds), and 2 percent to the observation area (14 beds). In addition, 27 percent of patients cared for in the main ED moved on to the observation unit prior to ED discharge.

When care areas were full, patients were directed to the waiting room until an appropriate bed was available. Patients remained in beds according to the LOS probability distributions associated with each ED care area stratified by admission status. For example, the LOS for patients discharged from the main ED followed a Weibull distribution with a mean of 4.8 hours. The mean LOS of main ED patients admitted to the hospital was 7.0 hours. The simulation was run for a 26-week period, after which results were collected. Next, verification and validation took place to ensure that the model operated correctly and the results accurately reflected the real-world system.

Verifying Proper Implementation

Verification (Step 5 in Figure 3) involves making certain that the DES

is implemented properly within the software (Sargent, 2004) and is primarily concerned with checking the model building (i.e., all inputs). For example, the verification step checks that LOS distributions for simulated patients within each ED care area match the distributions that were inputted into the model.

Validation (also in Step 5) moves this process a step further by determining that the model reflects the real-world system with a reasonable degree of accuracy (Sargent, 2004). Operational validity is particularly critical in healthcare DES models where simulated performance measures are compared to the real system. The DES model must describe performance of the system accurately to provide a baseline for comparing the effects of potential changes. The major validation measures for the ED DES created were bed occupancy (average 22.7 patients), waiting room occupancy (average 9.3 patients), waiting time distribution (average 2.1 hours), and patient volume (59,100 patients annually). The simulated measures were determined to reasonably match performance of the real ED system. Temporal patterns of bed and waiting room occupancy were also checked to ensure that the DES accurately accounted for variability over time.

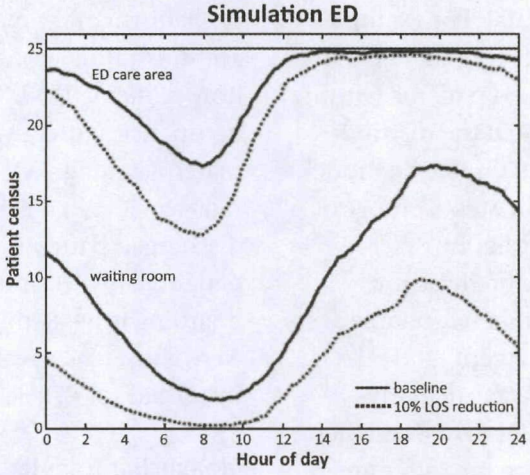
Testing Operational Scenarios

A validated DES may then be used to test different operational scenarios (Step 6 in Figure 3) and determine their effects on performance. A systematic approach to introducing changes to a DES model is recommended, which involves altering one component of the model at a time to assess the isolated effects of the change. This approach may be helpful in determining where changing operations or resources have the largest impact on performance. In our DES example, administrators wished to determine how reducing ED LOS would affect bed occupancy, patient waiting, and, ultimately, staffing. LOS distributions were reduced 10 percent across all patients, discharged patients only, and admitted patients only to simulate reductions in boarding time. Reductions of 10 percent were chosen as a realistic yet significant threshold for EDs to target.

Measuring Results Against Baselines

Performance measure results were then collected from the DES and compared to baseline runs (Step 7 in Figure 3). Simulating a 10 percent reduction in LOS across all ED patients showed improvements in patient flow that significantly affect the main ED care area and waiting room at different times. This effect is displayed in Figure 4, where patient census (i.e., occupancy) is averaged over all simulated days in the main ED care area and waiting room. The most substantial reductions in waiting time and waiting room occupancy occurred during evening hours (17:00 to 22:00). Significant effects on the main ED occupancy and staffing demands did not occur until overnight hours (02:00 to 08:00). Simulation results suggest that increasing ED efficiency will most significantly decrease delays experienced by evening arrivals and provide opportunities for cost savings by reducing staff overnight (see Step 7 in Figure 3). These results may be useful to administrators to quantify how gains in efficiency (in this example, reductions in LOS) translate

FIGURE 4
Timing of Reductions in ED Crowding



to improved patient access (satisfaction) and differences in capacity and staff utilization.

Making Recommendations

DES can help administrators validate gut feelings regarding where operational changes could be made (Step 8 in Figure 3). In our example, a strong anecdotal belief prevailed among leadership that reducing patient LOS could provide additional capacity in the ED and present opportunities for changes to the current staffing model. By modeling projected demand in the simulation environment, changes in patient wait time and census were observed over time. The example shows that staffing could be adjusted during the overnight shift. If the assumption that decreases in LOS across all shifts equate to staffing decreases during all shifts had been acted on, crowding would have worsened during peak ED hours. Other assumptions, such as adding physical capacity or staff, can be tested in the same way to identify where resources should be targeted to improve efficiency.

SUMMARY AND CONCLUSION

In a rapidly changing healthcare environment, administrators face pressure to make their organizations competitive through successful operational and strategic decision making. Rising to competition has proven difficult as the complexity of healthcare delivery does not often lend itself to clear, easy decision making. Part of this difficulty stems from the subjective information provided to administrators by frontline staff, who often see only one piece of the system in which they work. Although valuable, this information must be supplemented with objectivity-based tools to ensure that the true picture of the healthcare system is captured. One such tool that has been proven effective in other industries is DES.

DES is a data-driven method designed to show why a decision does or does not make the most sense among various options. It provides decision makers with a powerful supplement to current change management processes. By including those affected by a change in the creation of a DES model, the administrator can generate buy-in for a proposed initiative. DES can also be used to dispel certain myths in the clinical area, such as the need for more clinical space when not all space is being utilized or, as in the study example, that decreasing LOS can allow for staffing reductions at specific times. Through technological advances, DES software and capabilities have become increasingly attainable to healthcare decision makers. The low-cost, lower-risk decision support that DES provides can help leaders avoid costly, ineffective decisions and allows the user to target resources to areas where they will be the most effective in improving quality, safety, or financial metrics.

DES has proven highly effective in defense contracting, manufacturing, and business applications, but relatively limited evidence of reported cost reduction and impact in healthcare delivery is available, despite increased use in the healthcare arena (Jahangirian, Taylor, & Young, 2010; Young et al., 2009). The likely cause is not ineffective DES but rather lack of implementation of results and recommendations or, if implemented, lack of follow-up and reporting (due to an absence of resources, time, and mandate). Numerous DES applications are found in healthcare, but very few demonstrate a pre-/postintervention comparison. Providing this comparison highlights the importance of pairing DES with effective implementation and monitoring outcomes of interventions to prove value. Further research and documentation of implementation results are recommended to exemplify the usefulness of DES as a valuable tool for healthcare administrators.

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PRACTITIONER APPLICATION

Felix J. Bradbury, RN, ScD, FACHE, senior principal, Accenture Healthcare Analytics, Health & Public Services, Global Markets, Houston, Texas

Healthcare organizations are searching for new tools and techniques to maximize efficiencies across their delivery systems. Discrete event simulation (DES) modeling is one such tool, and a powerful one. DES allows bottlenecks in day-to-day workflows to be identified, improvements made, and solutions developed. Such insight into their day-to-day processes enables leaders to test incremental improvements and make informed decisions about which enhancements work and which do not.

As the authors point out, DES is well suited for application within a clinical environment where the workflows vary from area to area, as from emergency department to laboratory. But DES may also be applied in health plans, including Medicare, Medicaid, and commercial plans.

As an example, a national Medicare Advantage payer launched a DES project to improve its claims processing. At the time the DES project was initiated, this payer had approximately 250,000 covered lives.

As with many payers, this organization struggled with prior-authorizations turnaround times (TATs)—the number of calendar days required by a health plan to process a routine authorization request. Prior to March 2011, the plan's TATs were averaging approximately 12.5 days ($SD \pm 9.97$ days). While a TAT on routine authorizations of less than 14 days is considered compliant with regulatory requirements, it is far outside the industry norm of 1–3 days. As a result of the high turnaround times, the plan also received a significant volume of calls through its member services call queue from physicians' offices and members asking for updates on the status of their authorizations. The department's poor performance was a source of dissatisfaction

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